MATH441/MATH541
Applied Mathematics II: Complex Variables
Lecture slides 10: Cauchy's Integral Formula

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Lecture outline

1. Deformation Theorem
2. Extended Deformation Theorem
3. Cauchy's Integral formula
4. Formula for Higher Derivatives
5. Liouville Theorem
6. The Fundamental Theorem of Algebra

Deformation Theorem

In what follows we adopt the convention that a closed path is oriented *positively* if it is oriented counterclockwise.

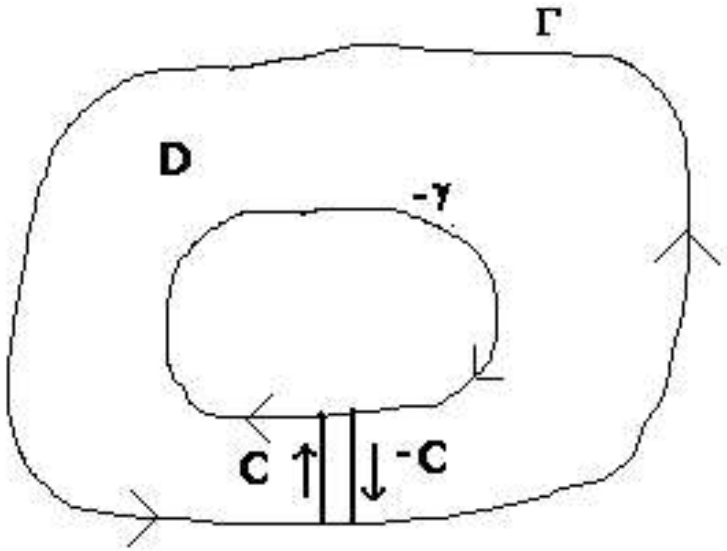
Theorem

Let Γ and γ be positively oriented closed paths, where γ is in the interior of Γ. Let f be differentiable on an open set containing both paths and the set of points between them. Then

$$\oint_{\Gamma} f(z)\,dz = \oint_{\gamma} f(z)\,dz$$

Remark: If one thinks of the curves as pieces of string, one curve can be continuously deformed into the other, moving only over points between the curves at which f is differentiable. Points at which f is not differentiable form a barrier to the deformation as far as the integral is concerned, so the region between the curves cannot contain any such points. In topology these curves are said to be *homotopic.*

Deformation Theorem 2



"Proof" :

$$\left[\int_{\Gamma}+\int_{C}+\int_{-\gamma}+\int_{-C}\right] f(z)\,dz = 0 \Rightarrow \oint_{\Gamma} f(z)\,dz = \oint_{\gamma} f(z)\,dz$$

Example 1

Example

Evaluate

$$\oint_\Gamma \frac{1}{z-a}dz$$

over any closed path containing a.
By the Deformation Theorem since $1/(z-a)$ is differentiable on both curves and the region between them, then representing $\gamma : z(t) = a + re^{i\theta} \quad 0 \leq \theta \leq 2\pi$ we have

$$\oint_\Gamma \frac{1}{z-a}dz = \oint_\gamma \frac{1}{z-a}dz = \int_0^{2\pi} \frac{1}{re^{i\theta}} ire^{i\theta} d\theta = 2\pi i$$

Thus we see that the integral on the left for any closed path enclosing a is equal to an integral which we can evaluate.

Extended Deformation Theorem

Instead of a single closed path γ enclosed by Γ we now consider any finite number of such paths enclosed by Γ.

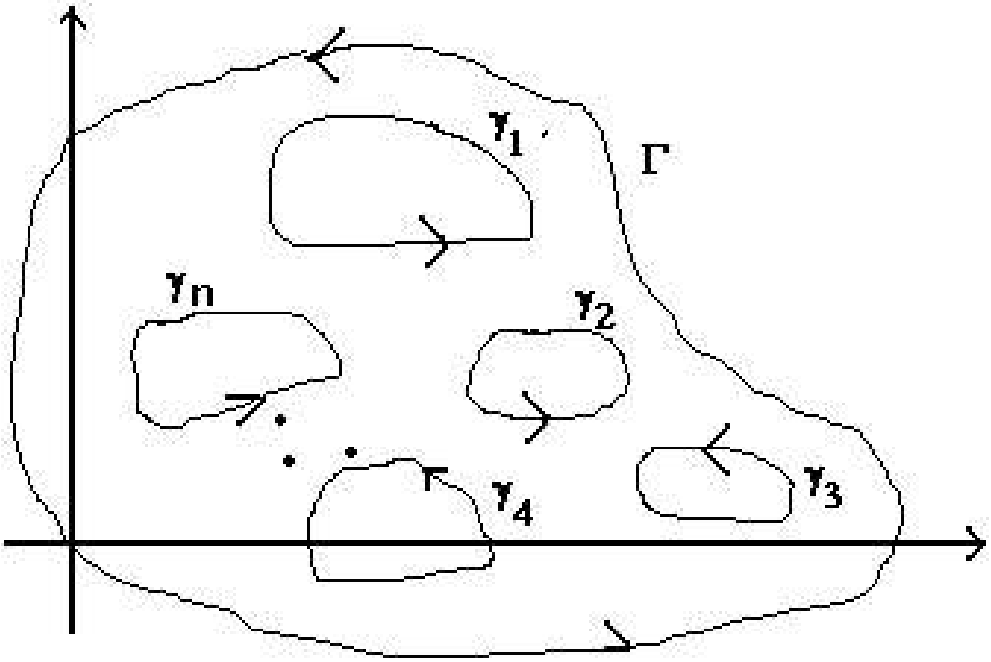
Theorem

Let Γ be a positively oriented closed path. Let $\gamma_1, \gamma_2, \ldots \gamma_n$ be positively oriented closed paths in the interior on Γ such that no two of Γ, $\gamma_1, \gamma_2, \ldots \gamma_n$ intersect, and no point interior to γ_j is in common with a point interior to another γ_k. Let f be differentiable on an open set containing Γ, each γ_j and all points that are both interior to Γ and exterior to each γ_j. Then

$$\oint_\Gamma f(z)dz = \sum_{j=1}^{n} \oint_{\gamma_j} f(z)\, dz$$

This is our earlier Deformation Theorem if $n = 1$.

Extended Deformation Theorem. 2.

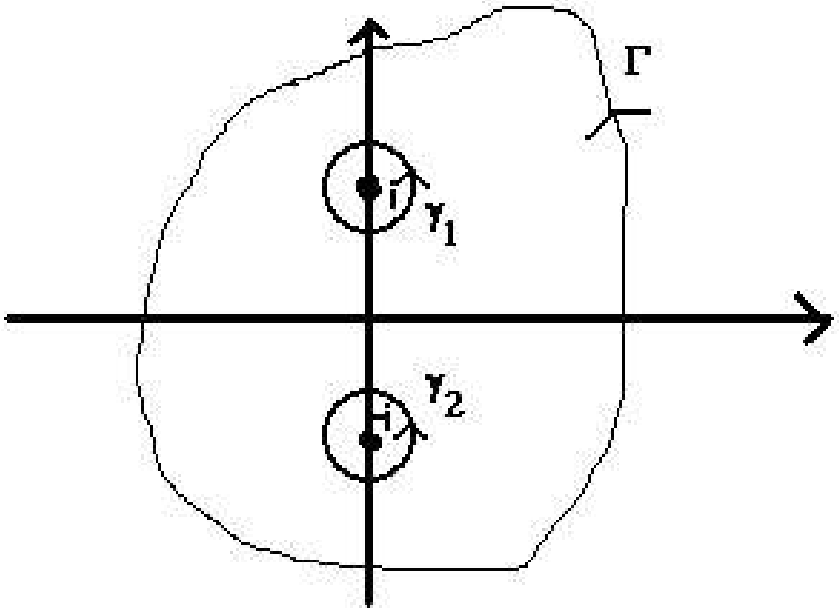


Example 2. Part 1.

Evaluate

$$\oint_\Gamma \frac{1}{z^2+1}\,dz,$$

where Γ is any closed path enclosing both i and $-i$.



We enclose i and $-i$ within two circles γ_1 and γ_2 of radius small enough so neither intersects the other and each lies entirely in the interior of Γ.

Example 2. Part 2.

Then $f(z) = 1/(z^2+1)$ is differentiable on Γ and the two circles and in the region between them. But

$$\frac{1}{z^2+1} = \frac{i}{2}\frac{1}{z+i} - \frac{i}{2}\frac{1}{z-i}$$

By the Extended Deformation Theorem

$$\oint_\Gamma \frac{1}{z^2+1}dz = \oint_{\gamma_1} \frac{1}{z^2+1}dz + \oint_{\gamma_2} \frac{1}{z^2+1}dz =$$

$$= \oint_{\gamma_1} \frac{i}{2}\frac{1}{z+i}dz - \oint_{\gamma_1} \frac{i}{2}\frac{1}{z-i}dz + \oint_{\gamma_2} \frac{i}{2}\frac{1}{z+i}dz - \oint_{\gamma_2} \frac{i}{2}\frac{1}{z-i}dz$$

But the first and fourth integrals are zero by Cauchy's Theorem, since the integrand is differentiable on and within the respective circles. The second and third integrals are $2\pi i$ by an earlier example. Hence

$$\oint_\Gamma \frac{1}{z^2+1}dz = -\frac{i}{2}(2\pi i) + \frac{i}{2}(2\pi i) = 0.$$

Cauchy Integral Formula. 1

Theorem

Cauchy Integral Formula Let f be differentiable on an open set G. Let Γ be a closed path enclosing only points of G. Then, for any z_0 in the interior of Γ

$$f(z_0) = \frac{1}{2\pi i} \oint_{\Gamma} \frac{f(z)}{z - z_0} dz$$

The Cauchy Integral Formula. 2

Let $f(z)$ be regular inside and on a simple closed contour C.
Let ζ be any (fixed) point in D the interior of C. The function $\frac{f(z)}{z-\zeta}$ is analytic and single-valued in D and on C except at $z=\zeta$.
Let C_r be the circle centered at ζ with radius $r>0$, where $C_r \subset D$ and both C and C_r are positively oriented with respect to their interiors.

$$\oint_C \frac{f(z)}{z-\zeta}dz = \oint_{C_r} \frac{f(z)}{z-\zeta}dz$$

Since $f(z)$ is continuous, we can take r so small that $|f(z)-f(\zeta)| < \in$ for z on C_r. $\Longrightarrow$

$$\oint_C \frac{f(z)}{z-\zeta}dz = \oint_{C_r} \frac{f(z)}{z-\zeta}dz =$$
$$= \oint_{C_r} \frac{f(\zeta)}{z-\zeta}dz + \oint_{C_r} \frac{f(z)-f(\zeta)}{z-\zeta}dz = 2\pi i f(\zeta) + \oint_{C_r} \frac{f(z)-f(\zeta)}{z-\zeta}dz$$

The Cauchy Integral Formula. 3

$$\left|\oint_C \frac{f(z)}{z-\zeta}dz - 2\pi i f(\zeta)\right| = \left|\oint_{C_r} \frac{f(z)-f(\zeta)}{z-\zeta}dz\right| \leq \frac{\in}{r}(2\pi r)$$

since z is on C_r so that $|z-\zeta| = r$.
Since the left hand side is independent of $\in$ and $\in > 0$ may be chosen arbitrarily small, the left hand side equals zero. Therefore

$$f(\zeta) = \frac{1}{2\pi i}\oint_C \frac{f(z)}{z-\zeta}dz$$

which is the Cauchy Integral Formula.

Example 3

Example

Evaluate

$$\oint_\Gamma \frac{e^{z^2}}{z-i}\,dz$$

for any closed path Γ whose graph does not pass through i.

We consider two cases: Case 1. The closed path Γ does not enclose i. Then $\frac{e^{z^2}}{z-i}$ is differentiable on Γ and in its interior. By Cauchy's Theorem the integral is zero.

Case 2. The closed path Γ encloses i. Now $f(z) = e^{z^2}$ is differentiable throughout the entire plane. By Cauchy's Integral Formula

$$\oint_\Gamma \frac{e^{z^2}}{z-i}\,dz = 2\pi i\, f(i) = 2\pi e^{-1}$$

Formula for Higher Derivatives

Theorem

Let f be differentiable on an open set G, and let Γ be a closed path in G enclosing only points of G. If z_0 is any point in the interior of Γ, then f has derivatives of all orders at z_0. Further, the nth derivative of f at z_0 is given by

$$f^{(n)}(z_0) = \frac{n!}{2\pi i} \oint_\Gamma \frac{f(z)}{(z - z_0)^{n+1}} dz$$

Remark: The integral above is exactly what we would get if we differentiated Cauchy's Integral Formula n times with respect to z_0 under the integral sign.

Example 4

Example

Evaluate

$$\oint_\Gamma \frac{e^{z^3}}{(z-i)^3}\,dz$$

where Γ is a closed path that does not pass through i.
Again we have two cases. If Γ does not enclose i , then the integral is 0 by Cauchy's Theorem. If Γ does enclose i, then let $f(z) = e^{z^3}$. The integral formula for higher derivatives gives

$$\oint_\Gamma \frac{e^{z^3}}{(z-i)^3}\,dz = \frac{2\pi i}{2!} f''(i) =$$

$$= \pi i \left[\left(6z + 9z^4\right) e^{z^3}\right]_{z=i} = \pi i\,(6i+9)\,e^{-i} =$$

$$= 3\pi i\,(3+2i)\,(\cos 1 - i \sin 1)$$

Corollaries

Corollary

Let f be differentiable in an open set G. Then at each point of G, f has derivatives of all orders.

Corollary

Bounds on Derivatives

Let f be differentiable on an open set G. Let z_0 be a point of G, and let the open disk or radius r about z_0 be in G. Suppose also that $|f(z)| \leq M$ for z on the circle of radius r about z_0. Then for any positive integer n,

$$\left|f^{(n)}(z_0)\right| \leq \frac{Mn!}{r^n}$$

Liouville Theorem

Theorem

If f is differentiable and bounded over the entire complex plane, the f is a constant function.

Proof: Suppose $|f(z)| \leq M \ \ \forall \ \ z$. Choose any point ζ and any positive number r. By the previous Corollary with $n = 1$, we have

$$|f'(\zeta)| \leq \frac{M}{r}$$

Since r may be chosen as large as desired, we see that $|f'(z)| = 0$. Therefore $f'(\zeta) = 0 \ \ \forall \ \zeta$. Therefore, $f(z)$ must be constant on any open disk about the origin, and hence constant for all z.

Remark: There is no result analogous to Liouville's theorem for real functions. For example, $\sin x$ is bounded and differentiable for all real x.

The Fundamental Theorem of Algebra

Liouville's Theorem provides a simple proof of the fundamental theorem of algebra, which states that a nonconstant polynomial with complex coefficients must have a complex root.

Theorem

Let $p(z)$ be a nonconstant polynomial with complex coefficients. Then for some complex number ζ, $p(\zeta) = 0$.

Proof: Suppose $p(z) \neq 0 \ \forall \ z$. Then $1/p(z)$ is differentiable $\forall \ z$. Since $|p(z)| \to \infty$ as $|z| \to \infty$, then for some R, $|p(z)| > 1$ if $|z| > R$. Hence $1/p(z)$ is bounded for $|z| > R$. But $1/|p(z)|$ is continuous on the compact disk $|z| \leq R$ and is therefore also bounded for $|z| \leq R$. Therefore $1/p(z)$ is a differentiable function which is bounded on the entire plane. By Liouville's Theorem $1/p(z)$ must be a constant. This implies that $p(z)$ is a constant function, which is a contradiction. Thus there must be some z for which $p(z) = 0$.

Thank you